

Comprehensive R&D Workflow and Execution Plan for the Equine Oracle SaaS MVP (Lowkey Consultants Ltd)

This document integrates the optimal R&D workflow template for a SaaS MVP with the specific technical and strategic details of the Horse Race Predictor application. It provides a complete roadmap from initial research and development through to post-launch optimization and iteration.

Part 1: Optimal R&D Workflow Template (The Build-Measure-Learn Loop)

The R&D process is structured around a continuous **Build-Measure-Learn (BML)** loop, adapted for the specific context of a SaaS MVP. The goal is **validated learning**—to quickly test core hypotheses about the market, user needs, and technical feasibility.

Phase 1: Research and Discovery (The "Learn" Stage)

Step	Focus Area	Key Activities	Deliverables
1.1 Market Validation	Problem-Solution Fit	Conduct customer interviews, analyze competitor offerings, and define the core problem to be solved.	Problem Statement, Target User Persona, Competitive Analysis Report
1.2 Core Hypothesis Definition	Value Proposition	Formulate a clear, testable hypothesis for the MVP's core value proposition.	MVP Core Hypothesis, Success Metrics (AARRR/Pirate Metrics)
1.3 Technical Feasibility (PoC)	Risk Assessment	Conduct a Proof of Concept (PoC) for the riskiest technical component (e.g., the predictive model).	PoC Report (confirming technical viability or identifying blockers)
1.4 Feature Prioritization	Scope Definition	Use MoSCoW to define the absolute minimum feature set for the MVP.	MVP Feature Backlog (prioritized), User Stories

Phase 2: Design and Prototyping (The "Build" Stage - Conceptual)

Step	Focus Area	Key Activities	Deliverables
2.1 User Flow Mapping	User Experience (UX)	Map the end-to-end journey of the user completing the core task.	User Flow Diagram
2.2 Low-Fidelity Prototyping	Interface Design	Create wireframes and mockups focusing on layout and information architecture.	Wireframes/Mockups
2.3 Usability Testing	Early Feedback	Test the low-fidelity prototype with target users.	Usability Test Report
2.4 Architectural Blueprint	Technical Design	Define the initial technology stack, database schema, and high-level system architecture.	System Architecture Diagram, Technology Stack Decision

Phase 3: Development and Implementation (The "Build" Stage - Execution)

Step	Focus Area	Key Activities	Deliverables
3.1 Infrastructure Setup	DevOps/CI/CD	Set up cloud infrastructure, version control, and a basic CI/CD pipeline.	Live Staging Environment, Automated Deployment Script
3.2 Core Feature Development	Coding	Implement the Must-Have features, adhering to clean code principles.	Functional MVP Codebase
3.3 Automated Testing	Quality Assurance (QA)	Write unit, integration, and end-	Comprehensive Test Suite,

		to-end tests for all core features.	Minimum 80% Code Coverage
3.4 Security and Compliance	Trust and Reliability	Implement basic security measures and ensure compliance with relevant regulations.	Initial Security Audit Report

Phase 4: Launch and Data Collection (The "Measure" Stage)

Step	Focus Area	Key Activities	Deliverables
4.1 Beta/Pilot Launch	Deployment	Deploy the MVP to the production environment and onboard the first cohort of users.	Live MVP
4.2 Analytics Implementation	Data Tracking	Integrate robust analytics tools to track the defined success metrics (AARRR).	Analytics Dashboard
4.3 User Feedback Loop	Qualitative Data	Set up channels for direct user feedback.	Qualitative Feedback Log
4.4 Data Analysis	Hypothesis Testing	Analyze the collected data to validate or invalidate the core hypothesis.	MVP Performance Report (with clear Go/No-Go recommendation for next iteration)

Part 2: Equine Oracle MVP - Technical and Execution Details

This section integrates the specific technical details of the Horse Race Predictor application into the R&D workflow.

A. Data Strategy and Model Development (Phase 1 & 3 Execution)

The core R&D challenge is the predictive model, which requires a robust data strategy.

1. Data Sources and Collection (Enhanced)

- **Primary Target:** Official TAB API (<https://api.tab.co.nz/affiliates/v1/>) for programmatic access, managed by `data_ingestion_pipeline.py` .
- **Secondary/Historical Data:** Web scraping **RaceBase.co.nz** (and potentially NZTR) for historical race results, managed by `racebase_scraper.py` .
- **Scraping Technology:** Python with `requests` and `BeautifulSoup4` (`racebase_scraper.py`).
- **API Ingestion:** Python with `requests` and `pandas` (`data_ingestion_pipeline.py`).
- **Raw Data Storage:** Initial storage in **CSV (Comma-Separated Values)** format (`racebase_historical_data.csv`), with each row representing a single horse's participation in a race.

2. Feature Engineering

The strategy focuses on creating high-impact features to support the predictive model:

- **Form-Based Features:** `win_streak` , `place_streak` , `last_3_starts_avg_position` , `win_percentage` .
- **Jockey and Trainer Performance:** `jockey_win_rate` , `trainer_win_rate` , `jockey_trainer_combo_win_rate` .
- **Track and Distance Specialization:** `horse_track_win_rate` , `horse_distance_win_rate` , `horse_track_condition_win_rate` .
- **Time-Based Features:** `days_since_last_race` .
- **Encoding: One-hot encoding** for categorical features (`jockey` , `trainer` , `track`).

3. Model Training and Performance

- **Target Variable (Binary):** Prediction of a **Top 3 Finish** (`is_top3`) for better relevance to Trifecta/First Four bets.
- **Target Variable (Ranking):** **Relevance Score** for horses within a race, used for LightGBM Ranker.
- **Models Trained (Binary):** Logistic Regression, Random Forest, Gradient Boosting, XGBoost, LightGBM. The trained models (`*.pkl` files) and the `scaler.pkl` are now available in the `backend/models/` directory.
- **Models Trained (Ranking):** **LightGBM Ranker** (`train_ranking_model.py`) for direct prediction of horse order.
- **Cross-Validation: TimeSeriesSplit** (5 folds) is used for realistic temporal evaluation.

- **Current Performance (Binary):** The **Logistic Regression** model, when trained on the new advanced features, achieved a **ROC-AUC of 0.7893**, a significant improvement over the baseline (0.5175). (Source: `model_training_report_20251102_062059.md`)
- **Ranking Model Status:** The LightGBM Ranker proof-of-concept was successful. The next step is to integrate the full feature engineering pipeline and the final processed data (`racebase_processed_data_final_v2.csv`) to train the production-ready ranking model, which is the focus of the **Equine Oracle R&D Sprint** (Phase 4.3 in the Integrated Project Plan).
- **Next Step:** The LightGBM Ranker was successfully trained in a proof-of-concept run. The next step is to integrate the full feature engineering pipeline and the final processed data (`racebase_processed_data_final_v2.csv`) to train the production-ready ranking model, which is the focus of the **Equine Oracle R&D Sprint** (Phase 4.3 in the Integrated Project Plan).
- **Key Feature Importance:** The most important predictor is `track_dist_avg_pos` (average historical performance at a specific track and distance), with an importance score of **0.3306** (Source: `model_training_report_20251102_062059.md`).

B. Application Architecture and Integration (Phase 2 & 3 Execution)

The application follows a three-tiered architecture: Data Layer, Core Intelligence Layer, and Service Layer.

1. Backend API (Service Layer)

- **Framework:** Flask-based REST API (`prediction_api.py`).
- **Authentication:** API key-based authentication (`X-API-Key`).
- **Key Endpoints:**
 - `GET /health` : Health check.
 - `POST /predict` : Accepts race data and returns predictions from multiple models (e.g., `gradient_boosting` , `random_forest`).
 - `GET /compare-models` : Detailed comparison of model performance metrics.
 - `GET /feature-importance` : Top 10 most important features.
- **Deployment:** Development via `python3 prediction_api.py` ; Production via `gunicorn` or **Docker/Kubernetes**.

2. Android Application (Service Layer - Client)

- **Project Overview:** A premium Android application for predicting New Zealand TAB horse race winners, branded as the **Equine Oracle**. It is the core product of **Lowkey**

Consultants Ltd.

- **Technology Stack:**
 - **Language:** Kotlin
 - **UI Framework:** Jetpack Compose (Modern declarative UI)
 - **Architecture:** MVVM (Model-View-ViewModel)
 - **DI:** Hilt
 - **HTTP Client:** Retrofit + OkHttp
- **Key Features:**
 - Authentication & Subscription (API key management, premium feature access control).
 - Real-time predictions with confidence scores.
 - Local caching, offline mode, and prediction history.
 - Premium features: Advanced analytics, custom alerts.
- **Integration:** The Android app uses `PredictionApiService.kt` and `PredictionRepository.kt` to communicate with the Flask API, with `PredictionViewModel.kt` managing the UI state.

Part 3: Post-MVP R&D Execution Plan

This plan addresses the subsequent phases of **training, optimization, debugging, and iteration** following the initial MVP launch.

Phase 5: Training and Onboarding

This phase focuses on knowledge transfer and establishing the feedback loop.

Focus Area	Internal Activities	External Activities (User Onboarding)	Key Deliverables
Product Knowledge	Deep-dive sessions for support, sales, and engineering teams.	Create a concise help center, FAQs, and video tutorials.	Internal Training Materials, Public Help Center
Feedback Loop	Establish a clear process for escalating user-reported bugs and feature requests.	Implement an in-app guided tour to ensure users complete the "Aha!" moment.	Support Escalation Matrix, User Onboarding Metrics

Tooling	Train the team on how to interpret the Analytics Dashboard.	Set up a dedicated channel (e.g., Slack, Discord) for direct communication with early adopters.	Analytics Training Session, Community Channel Setup
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Phase 6: Optimization and Debugging (Stabilization)

This phase is dedicated to improving the stability, performance, and user experience based on real-world usage data.

Focus Area	Activities	Metrics for Success	R&D Output
Debugging	Prioritize and resolve critical bugs reported by users and error tracking.	Mean Time To Resolution (MTTR), Bug Count Reduction	Stable Codebase (v1.0.1), Detailed Bug Report Log
Performance Optimization	Load testing to identify bottlenecks; optimize slow queries and code paths.	Response Time (P95), Server Utilization Reduction	Performance Audit Report, Infrastructure Scaling Plan
UX Optimization	Run small-scale A/B tests on key conversion points (e.g., sign-up flow).	Conversion Rate Increase, Drop-off Rate Decrease	A/B Test Results and Implementation Plan
Security Hardening	Conduct a more thorough security scan and penetration test.	Zero Critical Vulnerabilities, Security Patch Log	Updated Security Policy, Penetration Test Report

Phase 7: Next Iteration (MVP 2.0 Development)

The R&D team uses validated learning to define and build the next set of features, looping back to Phase 1 of the R&D Workflow.

Step	Description	Input from Previous Phases	Output for Next Iteration
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7.1 Hypothesis Refinement	Re-evaluate the core hypothesis based on MVP performance data and user feedback. Define the next most critical problem to solve.	MVP Performance Report, Qualitative Feedback Log	MVP 2.0 Core Hypothesis
7.2 Feature Definition	Define the next set of features that directly address the refined hypothesis.	A/B Test Results, User Stories from Feedback	MVP 2.0 Feature Backlog
7.3 Development Cycle	Execute the Build-Measure-Learn loop again for the MVP 2.0 feature set.	MVP 2.0 Feature Backlog, Stable Codebase (v1.0.1)	MVP 2.0 Release Candidate
7.4 Scaling Preparation	Plan for the infrastructure and organizational changes required to support a larger user base and the new feature set.	Infrastructure Scaling Plan, Performance Audit Report	Roadmap for Scaling to 10x Users

Future Enhancements (Beyond MVP 2.0)

C. Financial Projections and Business Strategy (Supporting Documentation)

The following key business and financial documents have been prepared to support the MVP's launch and funding strategy:

| Document | Key Insight | Status |

Integrated Project Plan	Outlines the 4-week sprint to secure funding (Callaghan Grant, Angel Seed Capital) and establish the full multi-entity legal structure (Holding Co., Lowkey Consultants Ltd, Lone Wolf Customs, Private Trust, Charitable Trust).	Complete
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3-Year Financial Projections	Projects EBITDA of over \$2M by Year 3 with a Net Profit Margin exceeding 83% , based on a dual B2C subscription and B2B data licensing model.	Complete
Funding Strategy Documents	Outlines the strategy for securing the Callaghan Grant and Angel Investment, with a focus on the high-margin, scalable nature of the SaaS model.	Complete
NZ Business Entity Options	Provides a summary of the best legal structures (e.g., LLC equivalent) for the New Zealand market.	Complete
Comprehensive Action Plan	Details the non-technical steps for the New Zealand app startup, including legal, banking, and market entry.	Complete

Future Enhancements (Beyond MVP 2.0)

- **Real-Time Data Integration:** Connect to live TAB data feeds.
- **Advanced Models:** Integrate deep learning models (LSTM, Transformers).
- **Ensemble Methods:** Combine predictions from multiple models.
- **Explainability:** Implement SHAP values for model interpretability.
- **A/B Testing:** Test different model versions in production.
- **Feedback Loop:** Collect user feedback to improve predictions.

References

- [1] The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses. Eric Ries.
- [2] The Build-Measure-Learn Feedback Loop. Itonics Innovation.
- [3] SaaS MVPs Done Right: Best Practices for Subscription-Based Startups. Alter Square.